

SHALINI SINGH SIROLA

GitHub | LinkedIn | shaliniSirola@gmail.com

OBJECTIVE

Computer Science Engineering student with a strong academic record (CGPA: 9.14) and a solid foundation in Data Structures and Algorithms, object-oriented programming, and software development. Passionate about data-driven solutions, machine learning, and backend development, with hands-on experience in the MERN stack. Quick learner with strong analytical and problem-solving skills, eager to apply technical knowledge to build scalable applications and solve real-world challenges.

EDUCATION

Dayananda Sagar College of Engineering, Bangalore

2023 – 2027

B.E. in Computer Science and Engineering

CGPA: 9.14 (Upto 6th Semester)

TECHNICAL SKILLS

Programming Languages: C++, C, Python

Databases: MySQL, MongoDB

Cloud Platforms: Google Cloud Platform

Python Libraries: Pandas, NumPy, PySpark, SQLAlchemy

Visualization: Power BI, Tableau, Matplotlib

Frontend: HTML, CSS, React

Backend: FastAPI, REST APIs, Node.js

Tools: Git, GitHub, Linux, Docker, Postman, VS Code

Relevant Coursework: DBMS, Data Structures, Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming

PROJECTS

Visit South Africa – Travel Website

[Code](#)

[Preview](#)

- Designed and developed a responsive travel website showcasing South African destinations and attractions.
- Implemented engaging layouts using HTML, CSS, and JavaScript to provide an immersive browsing experience.
- Focused on clean UI/UX principles while optimizing performance for desktop and mobile devices.

PlanGrid – ML-Based Material Demand Forecasting System

[Code](#)

[Preview](#)

- Developed a machine learning-powered forecasting platform that predicts material demand for power infrastructure projects using project parameters such as budget, location, tower type, substation type, and tax rates.
- Built a full-stack web application using **React.js**, **Flask**, **MongoDB**, and REST APIs, featuring an interactive dashboard for visualizing demand forecasts and supporting procurement planning.
- Implemented an **XGBoost Regression** model with data preprocessing and feature engineering to improve forecasting accuracy, enabling data-driven supply chain optimization and reducing risks of material shortages or overstocking.

Aura – AI Powered Desktop Assistant

[Code](#)

- Engineered a local AI desktop assistant using Python, FastAPI, LangChain, and Ollama with conversational memory, real-time streaming responses, and voice interaction.
- Built an intelligent Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and vector embeddings for semantic search across personal documents and project files.
- Integrated secure terminal execution, Whisper-based speech recognition, and a Tauri desktop interface to enhance AI-assisted productivity and automation.

ACHIEVEMENTS & CERTIFICATIONS

- **The Complete Full-Stack Web Development Bootcamp** by Dr. Angela Yu (Udemy)
- **Python Data Visualization: Matplotlib & Seaborn MasterClass** by Chris Bruehl (Udemy)
- **Google Cloud Career Launchpad Computing Foundations track** (Google Cloud)
- Solved 50+ problems on LeetCode focusing on Data Structures and Algorithms